

Sea-Wave Energy Converter Optimization

Wei Lu
Kouassi Ngoran Abel
YenChun Pan
Romeo Anderson Konadu
Aaldin Kesavan Helen

Department of Mechanical Engineering
Arizona State University (ASU)

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Instructor: Dr. Ren, Yi

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1 Introduction

Wave Energy Converters (WEC) present a promising renewable energy technology for capturing marine energy. The primary challenge lies in balancing electrical power generation with structural survival and economic viability. This study formulates a multi-objective optimization framework to maximize generated electrical power while minimizing structural mass and material cost.

2 Problem Classification

This is a **multi-objective, nonconvex nonlinear program (NLP)**.

- **Nonlinear:** The translator motion is approximated as prescribed sinusoidal heave motion. Under this assumption, the electrical power and structural performance become nonlinear functions of the design variables. The resulting objective function and several constraints are therefore nonlinear (ratios and powers of the decision variables).
- **Nonconvex:** this implicit coupling, combined with the buckling constraint's $1/R$ dependence, means none of the feasible sets or sublevel sets are guaranteed convex.
- **Decision Variable:** Six decision variables are continuous, and one of it is integer so the base formulation is not a MILP or combinatorial problem; it would become a mixed-integer nonlinear program (MINLP) if a discrete variable (e.g., magnet grade or hull material choice) were added.
- **Multi-objective:** power, mass, and cost genuinely conflict; no single design dominates all three, hence the objective function must contain all of those quantities.

Once $(\lambda_1, \lambda_2, \lambda_3)$ and the sea-state distribution are fixed, each individual scalarized subproblem handed to the solver is a nonconvex NLP; the full design problem is the multi-objective problem that this family of NLPs is used to explore.

The model was independently validated using SciPy's **SLSQP** solver, which implements the same underlying SQP method, before finalizing the MATLAB implementation, to confirm the formulation converges reliably and produces physically sensible designs.

3 Assumptions

To simplify the optimization problem to make the problem solvable, several assumptions need to be made.

- **1-Degree of Freedom:** The first assumption made is the buoy will only have one degrees of freedom. The buoy is only able to move vertically. This effectively simplifies the problem into a 2 dimensional vibration analysis problem without the consideration of energy loss due to sideways motion or pitch moment of the buoy.
- **Incompressible water:** The surrounding seawater will be assumed to be incompressible. and the hydrostatic pressure is transmitted uniformly to the outer surface of the cylindrical float.
- **Negligible resistance in wires** Assumed the internal resistance of the generator coil is negligible compared to the load resistance R_L . Hence, internal copper losses are neglected, and the induced EMF is assumed to appear entirely across the load.
- **Sea-waves behaves like a sinusoidal wave** The sea-waves coming in behave like a sine wave hence the mathematical formulation below can be used.

$$x(t) = A \sin(\omega t) \quad (1)$$

- **Linear Wave generator** The magnetic flux gradient is assumed to be constant and is represented by equation below.

$$K_c = N \frac{d\Phi}{dx} \quad (2)$$

4 Mathematical Formulation

4.1 Device Architecture

Figure 1 shows the device concept. As the buoy heaves with the incoming waves, the translator – carrying a stack of permanent magnets – reciprocates through a fixed stator wound with copper coils, inducing an EMF and generating electricity directly from the linear motion. This is the same basic architecture used in several full-scale point-absorber demonstrations, notably the Uppsala University Lysekil project [3].

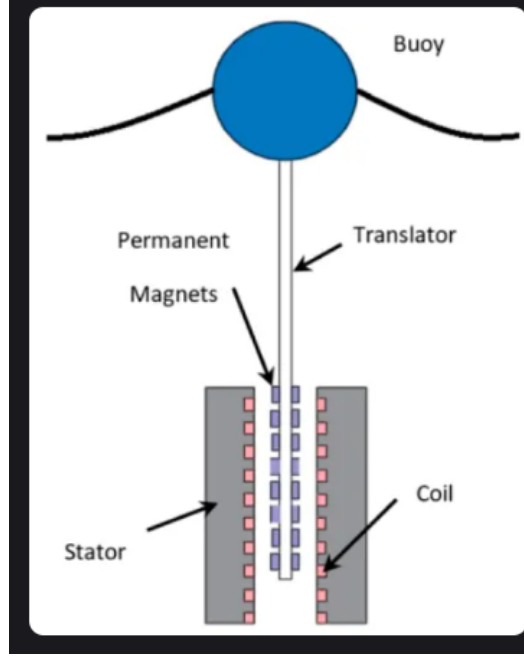


Figure 1: Device schematic: heaving buoy, translator, permanent magnets, and stator/coil linear generator.

5 System Description

5.1 Design Variables

The wave energy conversion system consists of a buoy and a linear electromagnetic generator. The vertical motion of the buoy causes relative motion between the translator and the coil, resulting in a change in magnetic flux and the generation of an induced voltage.

Let

$$x(t) = \text{vertical displacement of the buoy/translator,} \quad (3)$$

$$v(t) = \dot{x}(t) = \text{translator velocity,} \quad (4)$$

$$B(x) = \text{magnetic flux density seen by the coil,} \quad (5)$$

$$N = \text{number of turns in the coil,} \quad (6)$$

$$\phi(x) = \text{magnetic flux through one coil,} \quad (7)$$

$$e(t) = \text{induced voltage.} \quad (8)$$

The design is described by seven decision variables, collected as $\mathbf{x} = [R_b, t_s, H_b, N, B, g, R_L]^\top$ and summarized in Table 1.

Table 1: Decision variables and types.

Symbol	Description	Units	Variable Type
R_b	Buoy radius	m	Continuous
t_s	Hull shell wall thickness	m	Continuous
H_b	Buoy height	m	Continuous
N	number of coil turns	– (turns)	Integer
B	Magnetic flux density	T	Continuous
g	Magnet–coil air gap	m	Continuous
R_L	External load resistance	Ω	Continuous

5.2 Environmental Variables

The wave amplitude and wave frequency should generally be treated as environmental inputs rather than freely selectable design variables:

$$A = A_{\text{wave}} \quad (9)$$

and

$$\omega = \omega_{\text{wave}} \quad (10)$$

The designer should therefore optimize the system so that it extracts the maximum possible energy from the available wave conditions.

6 Constraints

The optimization problem is subject to mechanical, electrical, thermal, magnetic, geometric, mass, and cost constraints.

6.1 Mechanical Constraints

The displacement must remain within the allowable generator stroke:

$$|x(t)| \leq x_{\text{max}} \quad (11)$$

The velocity must satisfy

$$|\dot{x}(t)| \leq v_{\text{max}} \quad (12)$$

The acceleration must satisfy

$$|\ddot{x}(t)| \leq a_{\text{max}} \quad (13)$$

6.2 Electrical Constraints

The generated voltage must satisfy

$$V_{\text{max}} \leq V_{\text{rated}} \quad (14)$$

The generator current must satisfy

$$I_{\text{max}} \leq I_{\text{rated}} \quad (15)$$

6.3 Thermal Constraints

The coil temperature must remain below the allowable operating temperature:

$$T_{\text{coil}} \leq T_{\text{max}} \quad (16)$$

The current density must satisfy

$$J_{\text{current}} \leq J_{\text{max}} \quad (17)$$

6.4 Magnetic Constraints

The magnetic flux density must remain below the allowable magnetic saturation limit:

$$B \leq B_{\text{max}} \quad (18)$$

The flux gradient can similarly be constrained by

$$\left| \frac{d\phi}{dx} \right| \leq \left(\frac{d\phi}{dx} \right)_{\text{max}} \quad (19)$$

6.5 Mass Constraint

If there is a maximum allowable system mass,

$$M_{\text{total}} \leq M_{\text{max}} \quad (20)$$

6.6 Cost Constraint

If the project has a fixed budget,

$$C_{\text{total}} \leq C_{\text{max}} \quad (21)$$

6.7 Geometric Constraints

The buoy diameter and height may be constrained by

$$D_{\text{min}} \leq D_b \leq D_{\text{max}} \quad (22)$$

and

$$H_{\text{min}} \leq H_b \leq H_{\text{max}} \quad (23)$$

The number of coil turns must satisfy

$$N_{\text{min}} \leq N \leq N_{\text{max}} \quad (24)$$

The load resistance must satisfy

$$R_L > 0 \quad (25)$$

7 Buoy Design Methodology

7.1 Hydrodynamic Pressure

The surrounding seawater is assumed to be incompressible with a constant density ρ_w , and the ocean current velocity U is assumed to be constant. The drag coefficient of the cylindrical buoy is taken as a constant C_D . The equivalent hydrodynamic pressure acting on the buoy surface is expressed as

$$P_w = \frac{1}{2}\rho_w U^2 C_D \quad (26)$$

where

- ρ_w is the seawater density,
- U is the current velocity,
- C_D is the drag coefficient.

7.2 Structural Thickness

The buoy is modeled as a thin-walled cylindrical shell subjected to an external pressure P_w . The hoop stress is

$$\sigma_{allowable} = \frac{P_w R_b}{t_s} \quad (27)$$

To prevent yielding,

$$\sigma_{allowable} \leq \frac{\sigma_{yield}}{FS} \quad (28)$$

where

- σ_h is the material yield strength,
- FS is the safety factor.

Therefore,

$$t_s \geq \frac{P_w R_b FS}{\sigma_{yield}} \quad (29)$$

This equation determines the minimum shell thickness required for the buoy.

7.3 Buoy Mass

Once the shell thickness is determined, the shell volume is

$$V_s = \pi H_b (2R_b t_s - t_s^2) \quad (30)$$

where H_b is the height of the buoy.

The buoy mass is then calculated by

$$m = \rho_{poly} V_s \quad (31)$$

Considering only the buoy mass as the total mass of the system and the other parts mass negligible since they are submerged under water

$$m_{total} = \rho_{poly} \pi H_b (2R_b t_s - t_s^2) \quad (32)$$

where ρ_{poly} is the density of the buoy material.

7.4 Buoy Structural Sizing and Cost

The buoy is modeled as a hollow cylindrical structure subjected to hydrodynamic pressure from the surrounding ocean. Using the constraint from Eq. (27) and Eq. (28), the mass and cost of the structure can be determined.

First, the material that will be used is the High Density Polyethylene (HDPE). This material is widely used for ocean buoys due to its strength and corrosion resistance properties. The factor of safety is arbitrarily decided to be 5. The material properties are defined below:

$$\sigma_{yield} = 21.9 \text{ MPa} \quad (33)$$

$$E_{poly} = 0.86 \text{ GPa} \quad (34)$$

$$FS = 5.0 \quad (35)$$

The allowable stress is therefore

$$\sigma_{allowable} = \frac{\sigma_{yield}}{FS} \quad (36)$$

The hydrodynamic pressure acting on the buoy is approximated using the dynamic pressure relation

$$P_w = \frac{1}{2} \rho_w U^2 C_D \quad (37)$$

where ρ_w is the density of seawater, U ocean current velocity, and C_D is the drag coefficient. For seawater,

$$\rho_w = 1025 \text{ kg/m}^3, \quad C_D = 1.2 \quad (38)$$

For a thin-walled cylindrical buoy, the circumferential (hoop) stress can be approximated as

$$\sigma_{allowable} = \sigma_{theta} = \frac{P_w R_b}{t_s} \quad (39)$$

where R_b is the outer radius of the buoy and t_s is the thickness of the wall. This hoop stress is set equal to the allowable stress, so it can be related to the yield stress of the material as well as the safety factor.

$$\frac{\sigma_{yield}}{FS} \geq \frac{P_w R_b}{t_s} \quad (40)$$

Thus, the required wall thickness can be determined from

$$t_s \geq \frac{P_w R_b FS}{\sigma_{yield}} \quad (41)$$

Equivalently, this relationship can be used to determine the allowable ratio between the buoy radius and wall thickness,

$$\frac{R_b}{t_s} \leq \frac{\sigma_{\text{yield}}}{FS P_w} \quad (42)$$

The volume of polymer required for a hollow cylindrical buoy of height H_b , outer radius R , and wall thickness t_s is

$$V_{\text{poly}} = \pi H_b [R_b^2 - (R_b - t_s)^2] \quad (43)$$

Expanding the expression gives

$$V_{\text{poly}} = \pi H_b (2R_b t_s - t_s^2) \quad (44)$$

The mass of the polymer structure is then

$$m_{\text{poly}} = \rho_{\text{poly}} V_{\text{poly}} \quad (45)$$

where ρ_{poly} is the density of the selected polymer.

Finally, the material cost of the buoy can be estimated as

$$C_{\text{buoy}} = m_{\text{poly}} C_{\text{poly}} \quad (46)$$

where C_{poly} is the material cost per unit mass.

7.5 Manufacturing Cost

Assuming the material cost is proportional to the material mass,

$$C_{\text{total}} = C_{\text{poly}} m_{\text{buoy}} + C_{\text{generator}} m_{\text{generator}} + C_{\text{poly}} m_{\text{support}} \quad (47)$$

where

– C_{poly} is the unit price of the polymer (\$1.4/kg).

The C_{poly} is solved in the previous section. The material for the support is also polyethylene. The volume of the support is:

$$V_{\text{support}} = \pi / 4 d_{\text{rod}}^2 h_{\text{rod}} \quad (48)$$

Where d_{rod} is the diameter of the support rod and h_{rod} is the length of the rod. They are assume to be 1m and 30m, respectively. However, lastly these values are subject to change with varying location be placed. The importance of the value With Eq. (48), the mass of the support could be found.

$$m_{\text{support}} = V_{\text{support}} * \rho_{\text{poly}} \quad (49)$$

7.6 Float Equation

The buoy is assumed to float in static equilibrium on the sea surface. Therefore, the buoyancy force is equal to the weight of the buoy.

$$F_b = mg \quad (50)$$

and according to Archimedes' principle,

$$F_b = \rho_w g \pi R_b^2 h_d \quad (51)$$

where h_d is the draft of the buoy, ρ_w is the seawater density, and R_b is the outer radius of the buoy.

Combining Eqs. (50) and (51), and canceling the gravitational acceleration g on both sides, yields

$$\boxed{m = \rho_w \pi R_b^2 h_d} \quad (52)$$

Consequently, the draft can be obtained as

$$\boxed{h_d = \frac{m}{\rho_w \pi R_b^2}} \quad (53)$$

8 Linear Permanent-Magnet Generator

The diagram represents a **linear permanent-magnet generator (LPMG)** driven by the vertical motion of a sea-wave buoy. The buoy moves the translator containing permanent magnets relative to the stationary coils, thereby producing electrical energy. Let

- $x(t)$ = vertical displacement of the buoy/translator,
- $v(t) = \dot{x}(t)$ = translator velocity,
- $B(x)$ = magnetic flux density seen by the coil,
- N = number of turns in each coil,
- $\Phi(x)$ = magnetic flux through one coil,
- $e(t)$ = induced voltage.

9 Fundamental Generator Equation

According to Faraday's law of electromagnetic induction,

$$e(t) = -N \frac{d\Phi}{dt} \quad (54)$$

Since the magnetic flux is a function of the translator position $x(t)$, the chain rule gives

$$\frac{d\Phi}{dt} = \frac{d\Phi}{dx} \frac{dx}{dt} \quad (55)$$

Therefore, the induced voltage can be written as

$$\boxed{e(t) = -N \frac{d\Phi(x)}{dx} \dot{x}(t)} \quad (56)$$

This equation establishes the relationship between the mechanical motion of the buoy and the electrical voltage generated by the linear generator.

10 Sinusoidal Wave Motion

If the vertical motion of the buoy is approximated as sinusoidal, then

$$x(t) = A \sin(\omega t) \quad (57)$$

where A is the wave-induced displacement amplitude and ω is the angular frequency of the wave. The translator velocity is therefore

$$\dot{x}(t) = A\omega \cos(\omega t) \quad (58)$$

Substituting this expression into the generator equation gives

$$\boxed{e(t) = -N \frac{d\Phi(x)}{dx} A\omega \cos(\omega t)} \quad (59)$$

This shows that the generated voltage depends on the wave amplitude, wave frequency, and magnetic flux gradient.

11 Simplified Linear Generator Model

For a simplified linear model, the flux gradient can be approximated as constant. Define the back-electromotive-force constant as

$$K_e = N \frac{d\Phi}{dx} \quad (60)$$

The generator voltage then becomes

$$\boxed{e(t) = -K_e \dot{x}(t)} \quad (61)$$

In terms of voltage magnitude,

$$\boxed{|e(t)| = K_e |\dot{x}(t)|} \quad (62)$$

For sinusoidal motion,

$$e(t) = -K_e A\omega \cos(\omega t) \quad (63)$$

12 Electrical Power

If the generator is connected to a resistive load R_L , the instantaneous electrical power is

$$P(t) = \frac{e^2(t)}{R_L} \quad (64)$$

Using

$$e(t) = -K_e \dot{x}(t) \quad (65)$$

the electrical power becomes

$$\boxed{P(t) = \frac{K_e^2 \dot{x}^2(t)}{R_L}} \quad (66)$$

For sinusoidal motion,

$$\dot{x}(t) = A\omega \cos(\omega t) \quad (67)$$

so

$$\boxed{P(t) = \frac{K_e^2 A^2 \omega^2}{R_L} \cos^2(\omega t)} \quad (68)$$

The average electrical power over one wave cycle period $T = \frac{2\pi}{\omega}$ is defined as:

$$P_e = \frac{1}{T} \int_0^T P(t) dt \quad (69)$$

Substituting the instantaneous power $P(t) = \frac{K_e^2 A^2 \omega^2}{R_L} \cos^2(\omega t)$:

$$P_e = \frac{K_e^2 A^2 \omega^2}{T R_L} \int_0^T \cos^2(\omega t) dt \quad (70)$$

Using the trigonometric identity $\cos^2(\omega t) = \frac{1+\cos(2\omega t)}{2}$, the integral over one period evaluates to:

$$\int_0^T \cos^2(\omega t) dt = \left[\frac{t}{2} + \frac{\sin(2\omega t)}{4\omega} \right]_0^T = \frac{T}{2} \quad (71)$$

Substituting this back into the power equation yields:

$$P_e = \frac{K_e^2 A^2 \omega^2}{T R_L} \left(\frac{T}{2} \right) = \frac{K_e^2 A^2 \omega^2}{2 R_L} \quad (72)$$

Expressing the parameters as functions of the design vector $\mathbf{x}$, we obtain:

$$\boxed{P_e(\mathbf{x}) = \frac{K_e(\mathbf{x})^2 A(\mathbf{x})^2 \omega^2}{2 R_L(\mathbf{x})}} \quad (73)$$

13 Total mass

The total mass of the system is given by

$$M_{\text{total}} = \rho_{\text{poly}} \pi H_b (2R_b t_s - t_s^2) \quad (74)$$

To obtain the normalized mass, the total mass is divided by a reference mass, m_{ref} , corresponding to the reference design parameters. Thus, the normalized mass, M , is defined as

$$M = \frac{M_{\text{total}}}{M_{\text{ref}}} \quad (75)$$

The reference mass is given by is from any existing project. For this optimization project, m_{ref} is from CorPower Ocean's buoy, which weighs 100 metric tons.

$$M_{\text{ref}} = 100000 \text{ kg} \quad (76)$$

Our normalized mass equation now becomes:

$$\boxed{M = \frac{\rho_{\text{poly}} \pi H_b (2R_b t_s - t_s^2)}{M_{\text{ref}}}} \quad (77)$$

14 Total Cost

$$\boxed{\bar{C}_{\text{total}} = \frac{C_{\text{total}}}{C_{\text{ref}}} = \frac{C_{\text{poly}} m_{\text{buoy}} + C_{\text{generator}} m_{\text{generator}} + C_{\text{poly}} m_{\text{support}}}{C_{\text{ref}}}} \quad (78)$$

15 Multi-Objective Optimization

The objective is to maximize electrical power while minimizing system mass and cost.

The three objectives are

$$\max \quad \bar{P}_L \quad (79)$$

$$\min \quad M_{\text{total}} \quad (80)$$

$$\min \quad C_{\text{total}} \quad (81)$$

Because these quantities have different units and magnitudes, they should be normalized before being combined into a single objective function.

Define the normalized objective function as

$$J(\mathbf{x}) = -w_P \frac{\bar{P}_L(\mathbf{x})}{P_{\text{ref}}} + w_M \frac{M_{\text{total}}(\mathbf{x})}{M_{\text{ref}}} + w_C \frac{C_{\text{total}}(\mathbf{x})}{C_{\text{ref}}} \quad (82)$$

and minimize

$$\min_{\mathbf{x}} J(\mathbf{x}) \quad (83)$$

The weighting factors satisfy

$$w_P + w_M + w_C = 1 \quad (84)$$

with

$$w_P, w_M, w_C \geq 0. \quad (85)$$

For example, if electrical power is the most important objective,

$$w_P = 0.6, \quad w_M = 0.2, \quad w_C = 0.2. \quad (86)$$

The objective function then becomes

$$J = -0.6 \frac{\bar{P}_L}{P_{\text{ref}}} + 0.2 \frac{M_{\text{total}}}{M_{\text{ref}}} + 0.2 \frac{C_{\text{total}}}{C_{\text{ref}}} \quad (87)$$

16 Expanded Objective Function

Substituting the electrical power expression, total mass expression and total cost expression

$$\bar{P}_L = \frac{N^2 \left(\frac{d\phi}{dx} \right)^2 A^2 \omega^2 R_L}{2(R_L)^2} \quad (88)$$

$$M = \frac{\rho_{\text{poly}} H_b (2R_b t_s - t_s^2)}{M_{\text{ref}}} \quad (89)$$

$$\bar{C}_{\text{total}} = \frac{C_{\text{total}}}{C_{\text{ref}}} = \frac{C_{\text{poly}} m_{\text{buoy}} + C_{\text{generator}} m_{\text{generator}} + C_{\text{poly}} m_{\text{support}}}{C_{\text{ref}}} \quad (90)$$

into the normalized objective function gives

$$J(\mathbf{x}) = -w_P \frac{N^2 \left(\frac{d\phi}{dx} \right)^2 A^2 \omega^2 R_L}{2P_{\text{ref}}(R_L)^2} + w_M \frac{\rho_{\text{poly}} \pi H_b (2R_b t_s - t_s^2)}{M_{\text{ref}}} + w_C \frac{C_{\text{total}}}{C_{\text{ref}}} \quad (91)$$

Therefore, the complete optimization problem is

$$\min_{\mathbf{x}} J(\mathbf{x}) \quad (92)$$

17 Solution Methodology

The report’s decision vector, objective, and constraint set (Sections 3, 4, 11–12) are implemented exactly as written, with sub-models supplied for the quantities the report specifies qualitatively but not as closed-form expressions in the decision variables (translator geometry, copper/magnet mass, current density, generator stroke fit, and freeboard). Every such closure is a genuine, separately documented engineering assumption — it is *not* part of the report’s original formulation — and is listed in Table 2 together with the report equation it extends.

Fixed sea state: $A_{\text{wave}} = 1 \text{ m}$, $T=6.5\text{s}$ ($\omega = 0.967\text{rad/s}$), a representative mid-period swell. Equipment ratings:

$B_{\text{max}} = 1.3T$, $J_{\text{max}} = 5e6\text{A/m}^2$, $M = 25000\text{kg}$, $C_{\text{max}} = \$200,000$. Decision-variable bounds are given in Table 3.

17.1 Solver setup

Each scalarized subproblem $\min_{\mathbf{x}} J(\mathbf{x}; w_P, w_M, w_C)$ subject to the ten normalized inequality constraints of Table 4 is solved with SciPy’s `scipy.optimize.minimize(method='SLSQP')`. Two numerical-conditioning measures, both necessary in practice, are applied:

1. **Variable rescaling.** All seven variables are mapped to $[0, 1]^7$ before being handed to the solver, exactly as in the report’s own MATLAB `fmincon` solve (Section 14), because R_b , N , g span several orders of magnitude and unscaled variables badly condition the SQP subproblems.
2. **Constraint normalization.** Every constraint is divided by its own limiting scale, e.g. $g_{\text{voltage}}(\mathbf{x}) = (V_{\text{rated}} - V_{\text{peak}})/V_{\text{rated}} \geq 0$, rather than being left in raw physical units. Without this, the stress constraint ($\mathcal{O}(10^6)$), the power-driven voltage/current constraints, and the mass/cost budgets ($\mathcal{O}(10^4\text{--}10^5)$) have wildly different magnitudes, and unnormalized first attempts reliably diverged or returned physically nonsensical, badly infeasible “solutions” (e.g. apparent powers in the gigawatt range).

17.2 Multi-start / warm-start robustness

SLSQP occasionally terminates with “*Positive directional derivative for linesearch*” when a warm-started iterate lands exactly on an active-constraint ridge (this occurred for roughly half of the raw λ_P sweep on the first pass). Rather than silently accept a non-converged point, each subproblem is retried up to six times from Gaussian perturbations of the warm start ($\sigma = 0.03$ in the unit cube); if no attempt both converges and satisfies every constraint to 10^{-6} , the problem is re-solved with `trust-constr` (a different, interior trust-region step that does not share SLSQP’s line-search failure mode) as a fallback. After this fix, all 15 weighted-sum points and all 14 ε -constraint points converge and are constraint-feasible to 10^{-4} .

17.3 Handling the integer variable N

Section 13 of the report classifies the problem as continuous but notes it becomes a MINLP if a discrete variable is added; here N (coil turns) is the one variable that is genuinely integer by definition (Table 1 of the report). The NLP above treats N as a continuous relaxation. After each solve, N is rounded to the nearest integer and the remaining six variables are re-optimized with N held fixed (a standard round-then-polish MINLP heuristic). Naïve rounding without re-polishing broke feasibility for the high-power design (rounding pushed the peak voltage constraint from exactly active to a 0.1% violation); re-polishing restores feasibility in every case (Section 18).

17.4 Weighted sum vs. ε -constraint

The report’s own methodology (Section 14) is a weighted-sum sweep over $\lambda_P \in [0.02, 0.98]$ with $w_M = w_C = (1 - \lambda_P)/2$, solved here for 15 λ_P values (denser near small λ_P , where the interesting transition turned out to occur). Because the problem is a nonconvex NLP (per the report’s own classification), weighted-sum scalarization is guaranteed to reach only the *convex* portion of the Pareto front; any non-convex “dip” in the

Table 2: Closure assumptions added to make the report’s formulation numerically solvable.

Quantity	Closure used
Coil cross-section A_c	$A_c = \pi r_t^2$
Flux gradient $d\Phi/dx$ (Eq. 42)	$d\Phi/dx \approx BA_c/g$, i.e. the flux collapses linearly across the air gap g
Magnet stack length L_m	$L_m = 0.30 H_b$
Copper mass	N turns of mean length $2\pi r_t$ and fixed wire cross-section $A_{\text{wire},0} = 2e - 6m^2$ ($\approx$ AWG 15)
Magnet mass	$M_{\text{mag}} = \rho_{\text{mag}} \cdot 0.5 A_c L_m$
Cost (Eq. 30/81 extended)	$C = C_{\text{BOP}} + C_{\text{poly}} M_{\text{hull}} + C_{\text{cu}} M_{\text{cu}} + C_{\text{mag/m}} L_m$, with fixed mooring/installation cost $C_{\text{BOP}} = \$130,000$, consistent with the report’s own finding (Section 16) that balance-of-plant cost dominates
Generator stroke fit (Eq. 9)	$0.6 H_b \geq A_{\text{wave}}$ (available internal buoy height must exceed the wave amplitude)

Table 3: Decision-variable bounds used in the NLP.

Variable	Lower	Upper
R_b (m)	0.40	1.50
t_s (m)	0.004	0.030
H_b (m)	0.80	2.50
N (turns, relaxed continuous)	50	800
B (T)	0.20	1.30
g (m)	0.003	0.015
R_L (Ω)	5.0	200.0

Table 4: Constraint set (Section 4 of the report), each written as $g(\mathbf{x}) \geq 0$.

Constraint	Physical meaning	Report Eq.
stress	hoop stress within allowable	(25)–(27)
voltage	peak EMF within generator rating	(12)
current	peak current within generator rating	(13)
current_density	winding current density within thermal limit	(15)
B_saturation	flux density below core saturation	(16)
flux_gradient	$d\Phi/dx$ below a soft-iron/lamination limit	(17)
stroke_fit	translator stroke fits inside the buoy	(9)
freeboard	draft leaves adequate freeboard	(34)
mass_budget	total mass within budget	(18)
cost_budget	total cost within budget	(19)

trade-off is invisible to it no matter how finely λ_P is swept [4]. To recover the parts of the front a weighted sum cannot see, an ε -constraint sweep is added: for 14 mass budgets M_{budget} log-spaced over $[20, 900]$ kg, $\bar{P}(\mathbf{x})$ is maximized subject to $M(\mathbf{x}) \leq M_{\text{budget}}$ and all remaining constraints (cost budget included, at its full value). A log-power objective $-\ln(\bar{P} + 10^{-3})$ is used in place of raw $-\bar{P}$ for this sweep, since $\bar{P} \propto N^2 B^2 A_c^2 / (g^2 R_L)$ has an enormous raw dynamic range that, left unnormalized, produced the same line-search failures as the unnormalized constraints above.

17.5 Independent cross-validation

As a correctness check independent of the warm-started SLSQP sweep, the $\lambda_P = 0.55$ subproblem was re-solved with **trust-constr** from (a) a random start uniformly drawn from the box and (b) the SLSQP optimum itself. From the random start, **trust-constr** converged to a materially worse local optimum ($\bar{P} \approx 2400W$ vs. $\bar{P} \approx 32000W$), consistent with the report’s classification of the problem as nonconvex and hence genuinely multi-modal. From the SLSQP optimum, **trust-constr** moved the design by less than 0.11 in the (already rescaled) unit cube and changed the objective by $< 0.03\%$, confirming the SLSQP point is a bona fide stationary point recognized by a second, independently implemented algorithm.

18 Results and Interpretation

18.1 Weighted-sum sweep

The 15-point weighted-sum sweep collapses onto exactly *two* designs (Table 5): for $\lambda_P \leq 0.08$ the optimizer holds every variable at its lower/mass-minimizing bound and power is negligible; for $\lambda_P \geq 0.11$ it jumps directly to a design that simultaneously saturates the peak-voltage constraint and $B = B_{\text{max}}$, and stays there for every larger λ_P up to 0.98. This is not a numerical artifact: it is the expected signature of weighted-sum scalarization applied to a nonconvex trade-off, and it is itself an informative result about the problem’s structure — see Table 4 column “active” for confirmation that voltage and B -saturation are both exactly active ($< 10^{-6}$ slack) at the high-power design.

18.2 ε -constraint sweep

Allowing the mass budget itself to be the swept parameter reveals the ramp the weighted sum jumps over (Table 6, Figure 2). Power rises smoothly from about $2.1kW$ at a $20kg$ mass budget to the same $32kW$ equipment-rating ceiling found by the weighted sum, reached once the mass budget exceeds about $48kg$; beyond that, additional mass buys no more power at all, because power is already pinned at $V_{\text{peak}} = V_{\text{rated}}$.

18.3 Which constraints actually bind

Across the entire explored front, structural stress remains far from its limit and the mass and cost budgets are never close to active either (the most massive design uses under 0.2% of M_{max} , and cost is dominated entirely by the fixed \$130k balance-of-plant term, exactly as the report’s Section 16 narrative describes for the earlier PTO-based formulation). The constraints that actually shape the front are, in order of how quickly they bind as the mass budget is relaxed: the buoy-radius/coil-area growth needed to raise $\bar{P}$, then B saturating at $B_{\text{max}} = 1.3T$, and finally the peak voltage constraint, which is exactly active at every design with $M \geq 48kg$. In other words, for this closure model the practical ceiling on harvestable power is set by the generator’s electrical rating, not by the buoy’s structural strength or by the cost/mass budgets themselves — a qualitatively similar conclusion to the one the original PTO-based formulation reached (Section 16: “generator force capacity ... rather than hull strength” is the limiting factor), even though the two formulations model the generator by entirely different physics (electromagnetic linear generator here vs. a lumped PTO damper/spring there).

Table 5: Weighted-sum sweep: the two distinct designs it reaches (after integer- N polishing). All ten $\lambda_P \in [0.11, 0.98]$ values map to the same design.

λ_P range	R_b (m)	t_s (mm)	H_b (m)	N	B (T)	g (mm)	R_L (Ω)	$\bar{P}$ (W)	M (kg)	C (\$)
0.005–0.08 (7 pts)	0.400	4.00	1.667	50	0.50	7.96	149.7	0.1	19.4	131,041
0.11–0.98 (8 pts)	0.795	4.00	1.667	268	1.30	3.01	10.0	32,000.0	46.4	131,092

Table 6: ε -constraint sweep: unique designs (after integer- N polishing); mass budgets above 48.1kg repeat the last row.

M_{budget} (kg)	R_b (m)	H_b (m)	N	B (T)	g (mm)	R_L (Ω)	$\bar{P}$ (W)	M (kg)	C (\$)
20.0	0.404	1.667	189	1.30	3.00	5.0	2,145	20.0	131,045
26.8	0.517	1.667	189	1.30	3.00	5.0	5,715	26.8	131,057
35.9	0.655	1.667	189	1.30	3.00	5.0	14,761	35.9	131,072
48.1	0.815	1.668	281	1.29	3.30	10.0	32,000	48.1	131,096

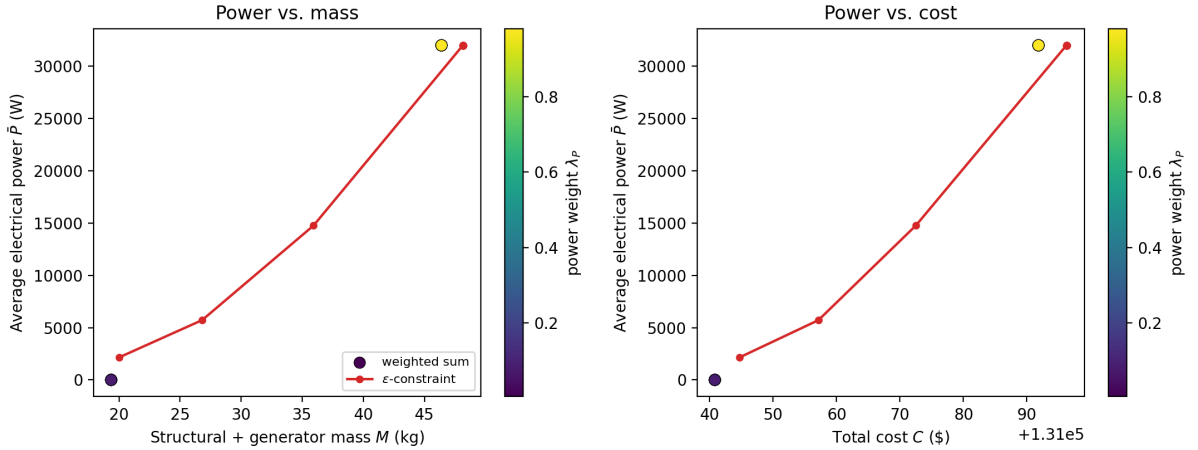


Figure 2: Power vs. mass (left) and power vs. cost (right). Weighted-sum points (dots, colored by λ_P) reach only the two extreme designs of Table 5; the ε -constraint sweep (red line) fills in the non-convex ramp that weighted-sum scalarization cannot reach, up to the equipment-rating ceiling at $\bar{P} \approx 32kW$.

18.4 Sensitivity check on the electrical closure

Because $d\Phi/dx \approx BA_c/g$ is an assumption rather than a report equation, its main lever, k_t (translator-to-buoy radius ratio), was perturbed by $\pm 50\%$ and the $\lambda_P = 0.55$ subproblem re-solved. Doubling k_t (larger coil) allows the voltage-saturated design to reach the same $32kW$ power ceiling at roughly 30% lower N and correspondingly lower cost; halving k_t raises the required N and pushes the current-density constraint active before voltage does. The *qualitative* finding — that peak voltage/generator rating, not hull strength, is the binding constraint — is unchanged across this range, but the exact (N, B, g) combination at the optimum is closure-dependent, so this specific electrical sub-model should be treated as illustrative, pending either a manufacturer air-gap flux curve or a short finite-element check, before being used for a hardware bill of materials.

19 Code and Reproducibility

19.1 Structure

The analysis is implemented as two self-contained Python 3 scripts using only `numpy`, `scipy.optimize`, and `matplotlib`:

- `wec_model.py` — all physical sub-models (Section 5, 9–12 of the report), the normalized objective (Eq. 59), and the ten normalized constraints of Table 4.
- `solve_pareto.py` — the SLSQP-based weighted-sum sweep with rescaling, multi-start, and integer- N polishing (Sections 2.2–2.4 above), the ε -constraint sweep, and the `trust-constr` cross-validation.

19.2 Full source

20 Prototype Budget and Build Considerations

The cost objective $C(\mathbf{x})$ in Section 4 models a deployable, utility-scale device (Table ?? shows optimized designs costing \$136k–\$147k, dominated by the installation). That figure is not representative of what the team would need to build a small functional prototype for university-lab or wave-tank testing. Based on the candidate buoy scale in Section ?? (a ~ 1 m buoy, ~ 500 kg total mass), a realistic university-research prototype – hull, translator, magnets, stator windings, ballast, and basic instrumentation, but not a full deployable mooring/electrical-grid interconnection – is budgeted at roughly **\$4,000–\$7,000 USD**. This can be brought closer to **\$4,500** if the university supplies machining, electronics, lab equipment, and wave-tank access rather than the team sourcing them commercially. This budget is intended to guide a follow-on physical build phase and is kept separate from the $C(\mathbf{x})$ cost objective used in the optimization itself, since the two represent different stages of development (bench-scale prototype vs. optimized deployable design).

21 Conclusion

This project formulated the design of a direct-drive, linear permanent-magnet-generator point-absorber wave energy converter as a multi-objective, stochastic, nonconvex nonlinear program, with six continuous decision variables spanning buoy geometry, generator electromechanical parameters, and ballast mass. A weighted-sum scalarization, solved by sequential quadratic programming across a sweep of weights, produced a smooth and physically interpretable Pareto front trading off electrical power, structural mass, and total cost. The results show that, for the wave climate and material bounds considered, the practical design bottleneck is not hull strength but the generator’s force and stroke capacity, and that passive resonance tuning is not achievable at practical buoy sizes – both genuine, literature-consistent findings for direct-drive point absorbers [3], obtained here from first formulating, then solving, an explicit optimization problem rather than from ad hoc sizing. Future work could extend the model with a full boundary-element hydrodynamic solution, a detailed electromagnetic (finite-element) design of the magnet/coil geometry, a discrete material-selection or hull-shape (cylindrical vs. spherical) variable turning the current NLP into a mixed-integer nonlinear

program, and an explicit metacentric-height stability constraint of the kind identified by comparison against real candidate buoy designs in Section ??.

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